

LAB EXPERIMENT RECORD

Set 12: Website Traffic Analysis

Department	B.Tech AIDS
Course Code / Course Name	DSA06 / Data Handling and Visualization
Experiment	Data Visualization using R
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Aim

To analyze website traffic data using a line chart, bar chart, and table in R, and to build an interactive R Shiny dashboard combining all three.

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%

PROGRAM 1: Line Chart of Daily Page Views

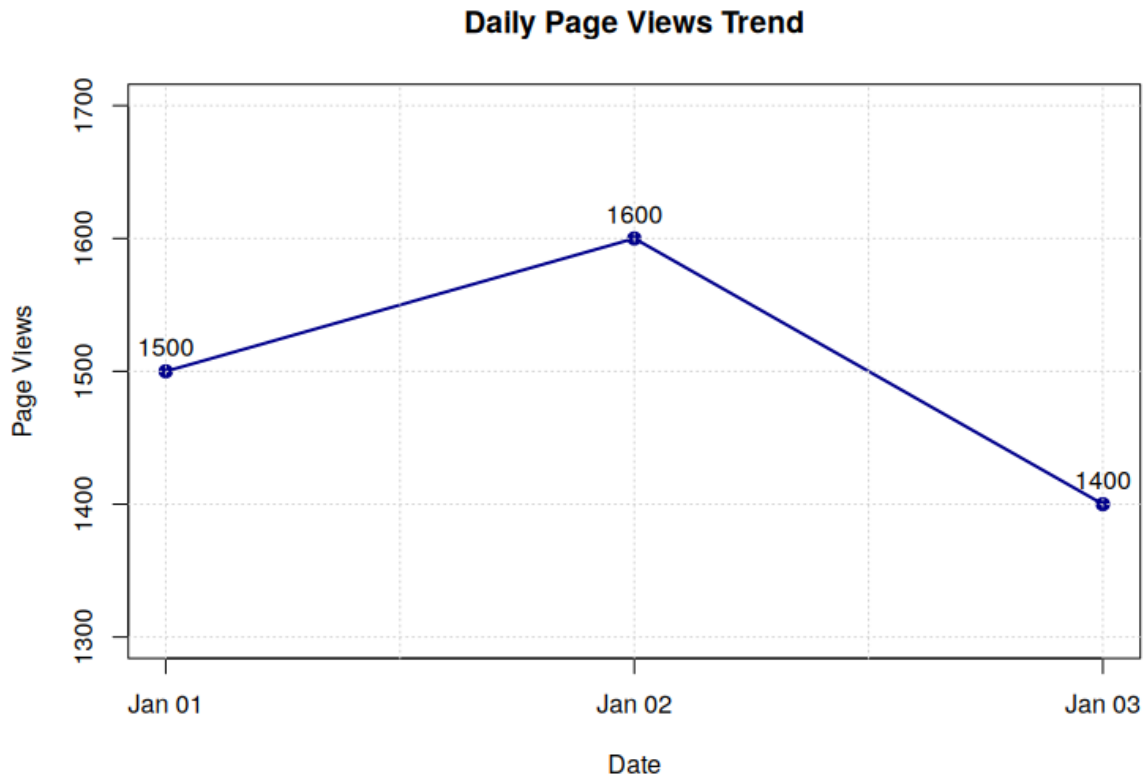
Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

```
# Create dataset for Set 12
traffic_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  Page_Views = c(1500, 1600, 1400),
  CTR = c(2.3, 2.7, 2.0)
)

print(traffic_data)

# 1. Line Chart of Daily Page Views
plot(traffic_data$Date, traffic_data$Page_Views,
     type = "o",
     col = "darkblue",
     lwd = 2,
     pch = 19,
     xaxt = "n",
     ylim = c(1300, 1700),
     xlab = "Date",
     ylab = "Page Views",
     main = "Daily Page Views Trend")
axis(1, at = traffic_data$Date, labels = format(traffic_data$Date, "%b %d"))
text(traffic_data$Date, traffic_data$Page_Views,
```

```
labels = traffic_data$Page_Views, pos = 3)  
grid()
```



Output 1: Daily Page Views Trend

Observation: Page views rise from 1500 on 01 January to the highest value of 1600 on 02 January, and then drop to the lowest value of 1400 on 03 January.

PROGRAM 2: Bar Chart of Top N Days by CTR

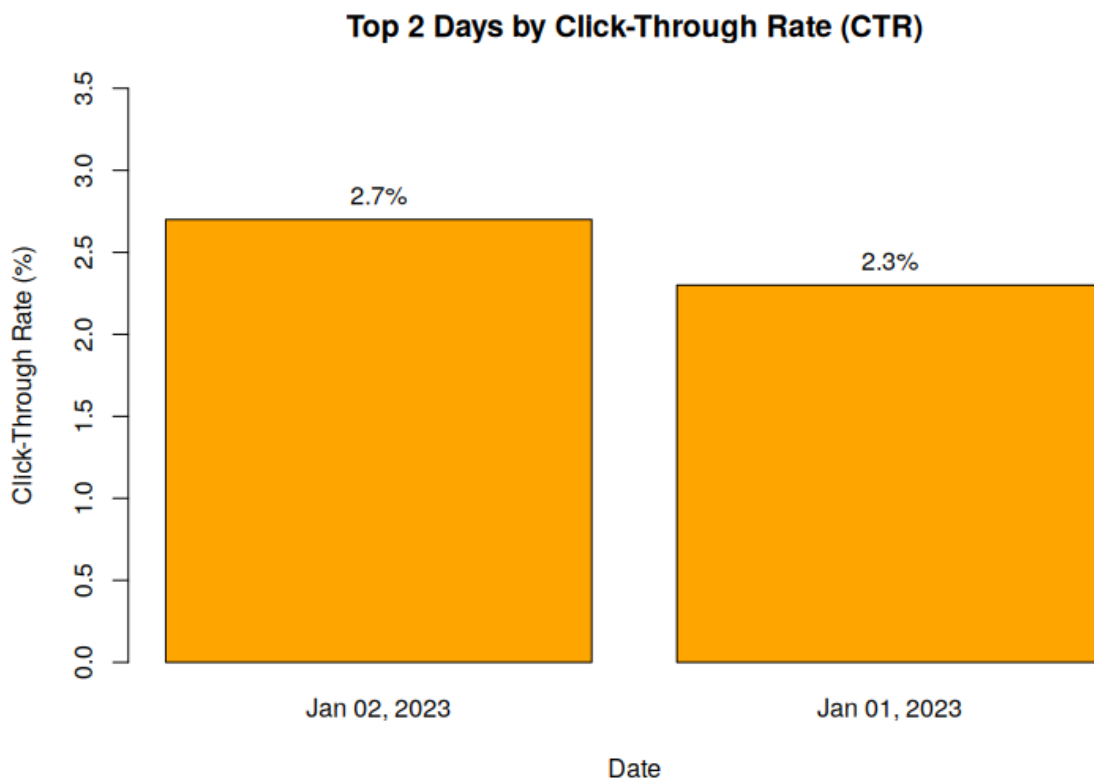
Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

```
# Create dataset for Set 12
traffic_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  Page_Views = c(1500, 1600, 1400),
  CTR = c(2.3, 2.7, 2.0)
)

print(traffic_data)

# Select the top N days by click-through rate
N <- 2
top_days <- head(traffic_data[order(-traffic_data$CTR), ], N)
print(top_days)

# 2. Bar Chart of Top N Days by CTR
bp <- barplot(top_days$CTR,
  names.arg = format(top_days$Date, "%b %d, %Y"),
  col = "orange",
  xlab = "Date",
  ylab = "Click-Through Rate (%)",
  main = paste("Top", N, "Days by Click-Through Rate (CTR)"),
  ylim = c(0, 3.5))
text(bp, top_days$CTR, labels = paste0(format(top_days$CTR, nsmall = 1), "%"), pos = 3)
```



Output 2: Top 2 Days by Click-Through Rate

Observation: With $N = 2$, the top days are 02 January (2.7%) and 01 January (2.3%). 03 January had the lowest CTR (2.0%) and is not in the top 2. The day with the highest page views also had the highest click-through rate. The value of N can be changed in the code.

PROGRAM 3: Table of Website Traffic Data

Build a table to show the website traffic data for each day. Label the table elements.

```
# Create dataset for Set 12
traffic_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  Page_Views = c(1500, 1600, 1400),
  CTR = c(2.3, 2.7, 2.0)
)

print(traffic_data)

# 3. Table of website traffic data
traffic_table <- traffic_data
traffic_table$CTR <- paste0(format(traffic_data$CTR, nsmall = 1), "%")
colnames(traffic_table) <- c("Date", "Page Views", "Click-through Rate")

cat("=== WEBSITE TRAFFIC TABLE ===\n")
print(traffic_table, row.names = FALSE)
```

```
=== WEBSITE TRAFFIC TABLE ===
> print(traffic_table, row.names = FALSE)
      Date Page Views Click-through Rate
2023-01-01      1500          2.3%
2023-01-02      1600          2.7%
2023-01-03      1400          2.0%
>
```

Output 3: Website Traffic Table

Observation: The table shows the page views and click-through rate for each day with clearly labelled columns. 02 January is the best day on both measures, and 03 January is the weakest.

PROGRAM 4: Interactive Dashboard for Website Traffic

Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of website traffic data.

Note: Tableau could not be installed on the system used for this experiment, so the interactive dashboard was developed in R using the Shiny package, with the same charts and interactive filters required in the question.

```
library(shiny)

# Dataset for Set 12
traffic_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  Page_Views = c(1500, 1600, 1400),
  CTR = c(2.3, 2.7, 2.0)
)
day_labels <- format(traffic_data$Date, "%b %d")

ui <- fluidPage(
  titlePanel("Website Traffic Dashboard"),
  sidebarLayout(
    sidebarPanel(width = 3,
      checkboxGroupInput("days", "Dates:", choices = day_labels, selected = day_labels),
      selectInput("topn", "Top N days by CTR:", choices = 1:3, selected = 2)
    ),
    mainPanel(width = 9,
      fluidRow(
        column(6, plotOutput("linePlot", height = "300px")),
        column(6, plotOutput("barPlot", height = "300px"))
      ),
      h4("Website Traffic Table"),
      tableOutput("trafficTable")
    )
  )
)

server <- function(input, output) {
  filtered <- reactive({ traffic_data[day_labels %in% input$days, ] })

  output$linePlot <- renderPlot({
    d <- filtered()
    plot(d$Date, d$Page_Views, type = "o", col = "darkblue", lwd = 2, pch = 19,
      xaxt = "n", ylim = c(1300, 1700), xlab = "Date", ylab = "Page Views",
      main = "Daily Page Views Trend")
    axis(1, at = d$Date, labels = format(d$Date, "%b %d"))
    grid()
  }, width = 420, height = 300, res = 96)

  output$barPlot <- renderPlot({
    d <- filtered()
    top <- head(d[order(-d$CTR), ], as.numeric(input$topn))
    barplot(top$CTR, names.arg = format(top$Date, "%b %d"), col = "orange",
      ylim = c(0, 3.5), xlab = "Date", ylab = "Click-Through Rate (%)",
      main = paste("Top", nrow(top), "Days by CTR"))
  }, width = 420, height = 300, res = 96)
}
```

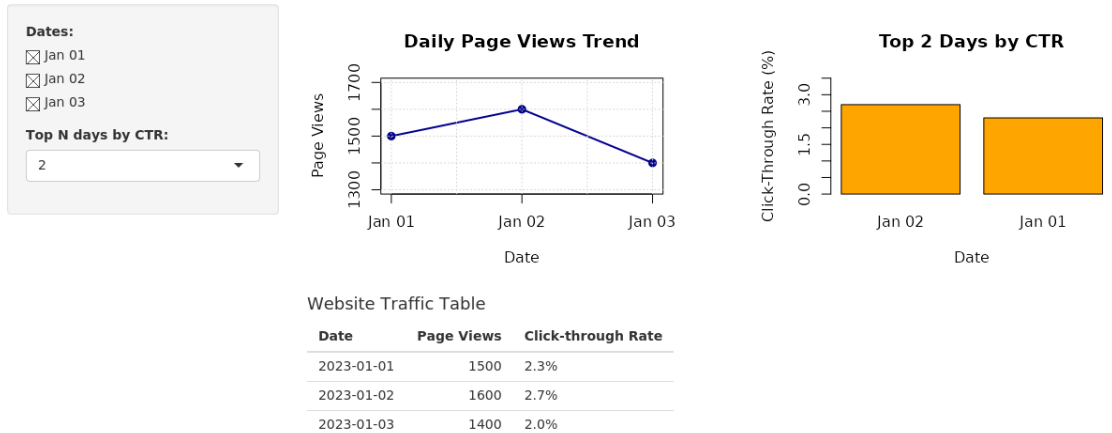
```

output$trafficTable <- renderTable({
  d <- filtered()
  d$Date <- format(d$Date, "%Y-%m-%d")
  d$CTR <- paste0(format(d$CTR, nsmall = 1), "%")
  colnames(d) <- c("Date", "Page Views", "Click-through Rate")
  d
}, digits = 0)
}

shinyApp(ui = ui, server = server)

```

Website Traffic Dashboard



Output 4: Website Traffic Interactive Dashboard (R Shiny)

Observation: The dashboard shows the page-view trend, the top N days by CTR, and the traffic table together. The Dates filter selects which days are shown, and the Top N box changes how many days appear in the bar chart. With N = 2, the top days are 02 January and 01 January.

Procedure

1. Create the website traffic data frame with Date, Page Views, and CTR.
2. Use `plot()` with `type = "o"` and `axis()` to draw the page-view line chart with date labels.
3. Sort the data by CTR using `order()`, select the top N days with `head()`, and draw the bar chart using `barplot()`.
4. Format the CTR values, rename the columns with `colnames()`, and print the labelled table.
5. Install and load the shiny package, and create the dashboard layout using `fluidPage()`, sidebar filters, and output areas for the charts and table.
6. Write the server function to filter the data with `reactive()` and redraw the outputs, then run the dashboard using `shinyApp()` and test the filters.

Result

The website traffic data was successfully visualized using a line chart, bar chart, and table in R, and an interactive R Shiny dashboard was built combining all three views.

Key R Functions Used

Function	Purpose
<code>as.Date()</code>	Converts text values into dates.
<code>plot()</code> / <code>axis()</code>	Creates the line chart with date labels.
<code>order()</code> / <code>head()</code>	Sorts days by CTR and selects the top N.
<code>barplot()</code>	Creates the CTR bar chart.
<code>text()</code>	Adds value labels to the charts.
<code>colnames()</code> / <code>print()</code>	Labels and displays the traffic table.
<code>fluidPage()</code> / <code>checkboxGroupInput()</code>	Creates the dashboard layout and filters.
<code>reactive()</code> / <code>renderPlot()</code>	Filters the data and redraws the charts.
<code>renderTable()</code>	Displays the filtered table on the dashboard.
<code>shinyApp()</code>	Runs the interactive dashboard.